
[Assessing Environmental Pressures on Dahan River Artificial Wetlands Using GIS: A Spatial Analysis Based on Land-Use Pollution Potential]

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1. INTRODUCTION

Under the dual pressures of global climate change and rapid urbanization, traditional gray-base facilities, which rely on high energy consumption and centralized treatment, are no longer sufficient to address the complex challenges of urban water environments. Therefore, the adoption of nature-based solutions and green infrastructure has become a core strategy for modern water conservancy engineering and urban planning. For the purposes of this study, it is essential to first establish the precise legal and environmental engineering status of constructed wetlands. This research is based on the legislative spirit of the Ministry of the Interior's Wetland Conservation Act, clearly defining constructed wetlands as an indispensable green infrastructure in contemporary urban river management and water resource management systems. According to the official definition and the Ministry of Environment's water quality protection policy, the core positioning of constructed wetlands transcends traditional landscaping and recreational functions; they simulate natural ecosystems, utilizing the complex physical, chemical, and biological synergies of vegetation, water bodies, and substrate networks to efficiently intercept and purify untreated domestic sewage and surface runoff rich in non-point source pollutants. Furthermore, under the legal principle of "No Net Loss," constructed wetlands also play a crucial role as a hub for ecological compensation in urban development. Therefore, their spatial location and layout must be closely coupled and linked with the confluence nodes of urban sewer systems, the development trajectory of land use zoning, and the water conservation and flood retention needs of riparian corridors. The above discussion not only establishes the multiple values of constructed wetlands in water resource management and urban development but also echoes the macro-vision of sustainable development in contemporary national policies. However, to accurately assess the specific benefits of constructed wetlands in improving urban river water quality and to formulate subsequent watershed management strategies, rigorous, objective, and legally valid water quality monitoring and pollution assessment indicators are essential.

2. STUDY AREA

The study area focuses on the Dahan River corridor in New Taipei City, specifically targeting eight major artificial wetlands: Daniaopi, Xinhai, Chenglin, Lujiaoxi, Fuzhou, and Huajiang. This massive ecological frontline faces high-density industrial and residential districts, making it an ideal field for assessing the conflict between green infrastructure and urban wastewater discharge

3. DATA COLLECTION

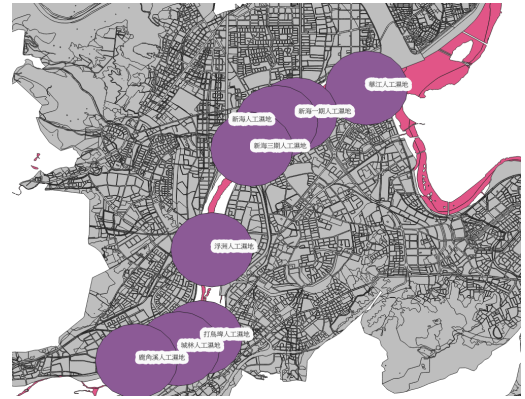
This research utilized the "Township and District Boundary" spatial data from the Ministry of Interior and the highly detailed "Urban Planning Land Use Zoning" map from the New Taipei City Government. The geographic coordinate points for the eight wetlands were initially plotted using Google Maps. To ensure the accuracy of geometric and area calculations, all spatial data layers were integrated and transformed into the TWD97 (EPSG:3826) coordinate reference system.

4. METHODS

This study employed QGIS as the spatial data processing platform. The core methodology consists of the following key steps:

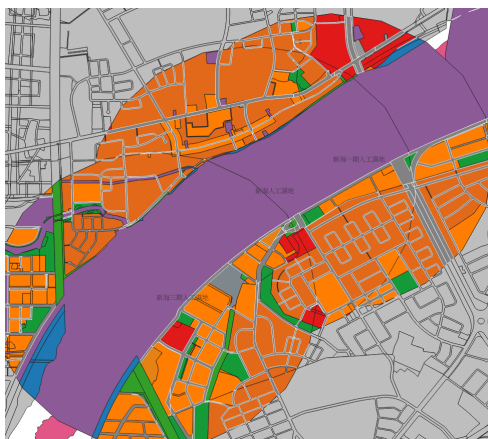
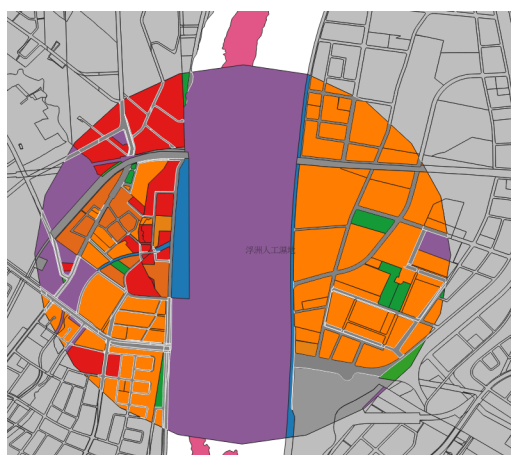
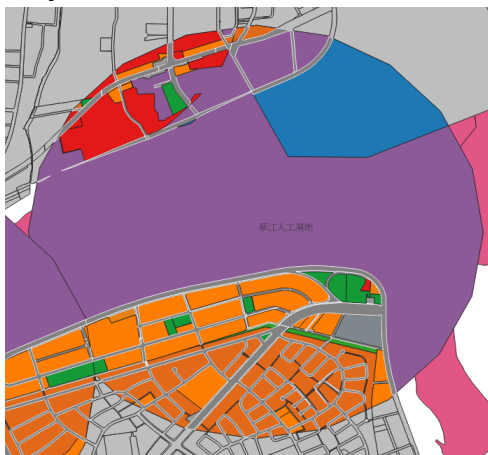
1. **Data Standardization and Point Mapping:** Open-source land use zoning maps from the government were imported into the system and uniformly converted to the TWD97 spatial coordinate system. The geometric center points of the 8 constructed wetlands were accurately positioned.
 2. **Setting the Potential Pollution Interception Zone (Buffer):** Centered on the wetlands, an absolute distance buffer of 1 kilometer was expanded outward to establish 8 independent environmental pressure diagnostic zones.
 3. **Spatial Intersection:** Using the 1-kilometer buffers as clipping geometries, complex land use zoning maps of New Taipei City were overlaid and intersected. This extracted the precise physical area and attribute data for every land use parcel within the buffers.
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step1



step2

step3



To transform complex land use zoning into specific pollution pressure indicators, this study reclassified the attribute data into five major environmental pressure indicators based on "pollution source characteristics" and "surface permeability":

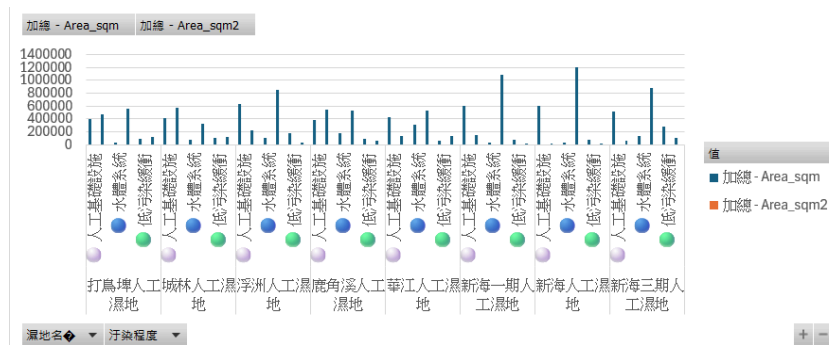
- **High Pollution Potential:** e.g., industrial zones and gas stations, posing threats of high-concentration heavy metals and chemical wastewater.
- **Moderate Pollution Potential:** e.g., high-density residential and commercial areas, which are the primary sources of massive daily domestic sewage.
- **Low Pollution Buffer:** e.g., agricultural zones and green spaces, possessing natural vegetation filtration and water retention capabilities.
- **Water Systems:** Existing rivers and water bodies in the vicinity.
- **Artificial Infrastructure:** e.g., roads, government agencies, and schools. These are entirely impermeable surfaces that accelerate surface runoff and pollution wash-off during rainfall.

5.RESULTS AND DISCUSSIONS

Through the quantified area data generated by GIS, this study conducted further statistical and visual analysis, revealing the following key findings:

5.1 Overall Environmental Pressure Assessment

After converting the areas of the five indicators into a 100% stacked bar chart, the results showed that, excluding the Huajiang wetland, the total pollution sources (High + Moderate Pollution Potential) for the other 7 wetlands all exceeded the **50% pollution pressure warning line**. The situation is most severe at the "Daniaopi Constructed Wetland," where over 60% of the land within its 1km buffer belongs to high and moderate pollution sources. This indicates that the wetland is severely caught between industrial and residential zones, bearing a massive environmental load.

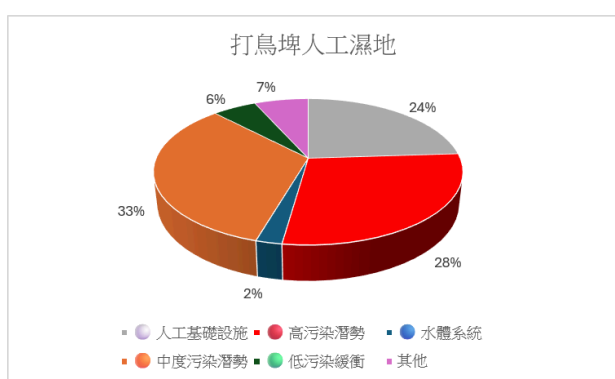
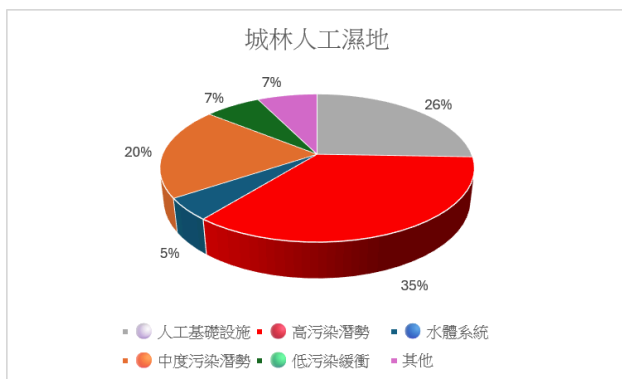
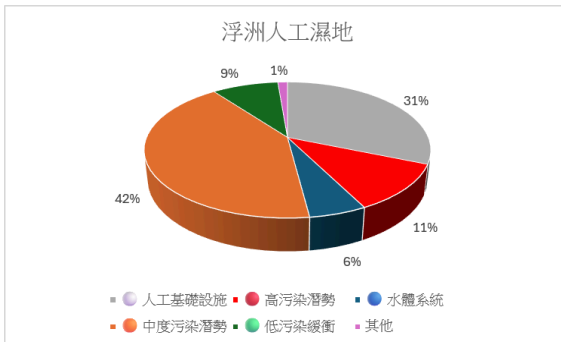
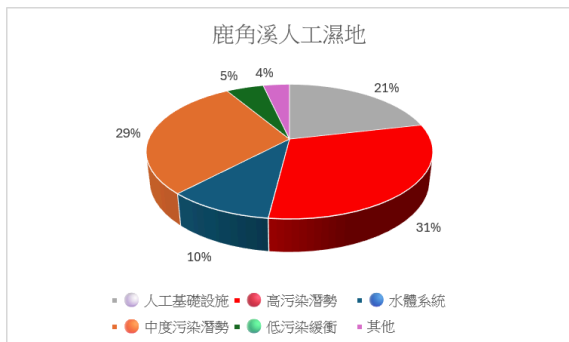
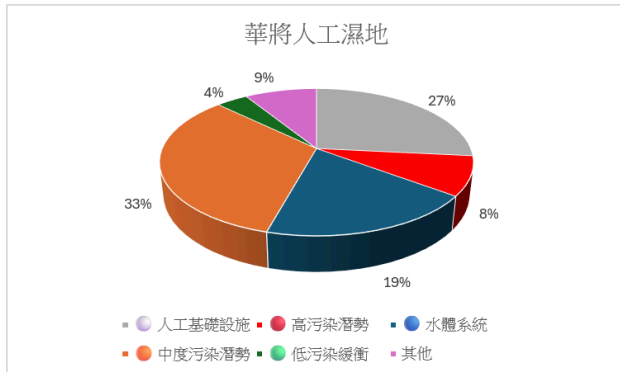
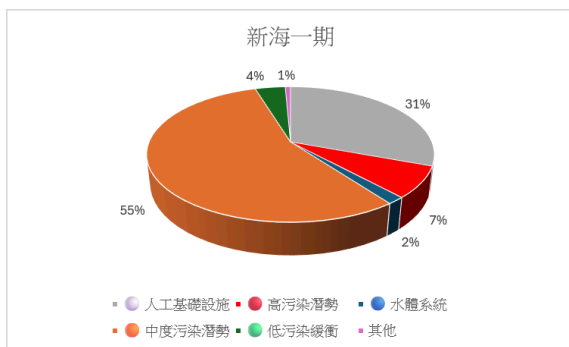
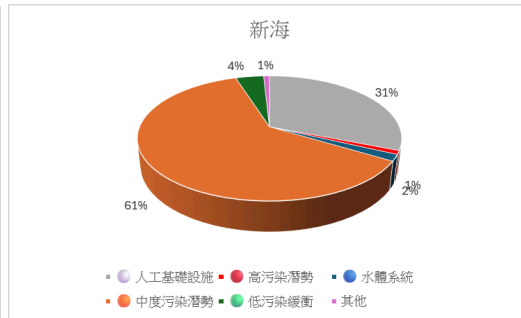
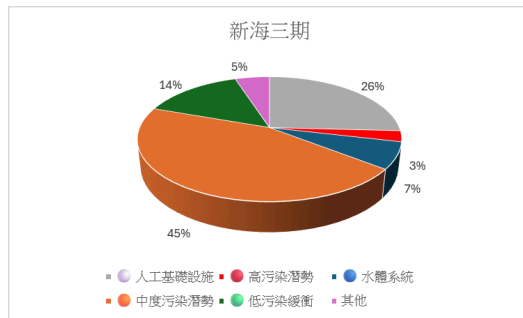


5.2 Spatial Differentiation of Pollution Structures

By further expanding each indicator into a side-by-side bar chart, this study confirmed that the "pollution structures" faced by each wetland differ significantly:

- **Industrial Pollution Frontline:** The "High Pollution Potential (Red)" indicator is exceptionally prominent around the Chenglin and Lujiaoxi wetlands, indicating that their primary interception targets are potential chemical and heavy metal wastewater discharged from industrial zones.
- **Domestic Sewage Hotspots:** The high pollution potential around Xinhai Phases I to III is extremely low, but the "Moderate Pollution Potential (Orange)" proportion is exceedingly high. This reflects that

their greatest challenge is the massive volume of domestic sewage, which is highly prone to causing water eutrophication.



6. CONCLUSION

This study focuses on eight artificial wetlands along the Dahan River. Through GIS 1-kilometer buffer zone analysis and land use reclassification, it analyzes the spatial pressure structure surrounding each wetland. It's important to note that there is currently **no single, universally accepted official comprehensive indicator that can directly convert the land use data into absolute pollution levels.**

Therefore, this study does not use fixed percentage thresholds to directly classify each wetland as "severely polluted" or "moderately polluted," but instead adopts a relative environmental pressure ranking method to present the factual differences faced by each artificial wetland under current land use conditions. **To avoid drawing conclusions based solely on a single percentage threshold, this study uses three indicators for interpretation.**

The first is the development stress index, calculated as follows:

$$\text{Development Stress Index} = \text{High Pollution Potential} + \text{Moderate Pollution Potential} + \text{Artificial Infrastructure}$$

If sorted using this indicator, the results are:

Xinhai Phase I Artificial Wetland: 1,826,782
Xinhai Artificial Wetland: 1,824,993
Fuzhou Artificial Wetland: 1,714,834
Lujiaoxi Artificial Wetland: 1,454,265
Xinhai Phase III Artificial Wetland: 1,452,889
Daniaopi Artificial Wetland: 1,439,331
Chenglin Artificial Wetland: 1,312,413
Huajiang Artificial Wetland: 1,086,444

This indicates that in terms of "total development-oriented land use," Xinhai Phase I and the area surrounding the Xinhai Artificial Wetland face the greatest development pressure.

The second is the ecological buffer index, calculated as follows:

$$\text{Ecological Buffer Index} = \text{Low Pollution Buffer} + \text{Water System}$$

If sorted using this indicator, the results are:

Xinhai Phase III Artificial Wetland: 423,929
Huajiang Artificial Wetland: 366,062
Fuzhou Artificial Wetland: 292,644
Lujiaoxi Artificial Wetland: 269,713
Chenglin Artificial Wetland: 189,352
Daniaopi Artificial Wetland: 130,806
Xinhai Artificial Wetland: 112,402
Xinhai Phase I Artificial Wetland: 111,708

This indicates that Xinhai Phase III and Huajiang Artificial Wetlands retain a relatively large number of water bodies and low-pollution buffer spaces, thus possessing relatively good ecological regulation conditions.

The third is the net environmental stress value, calculated as follows:

$$\text{Net Environmental Stress Value} = \text{Development Stress Index} - \text{Ecological Buffer Index}$$

Sorted by net environmental pressure value, the results are as follows:

Xinhai Phase I Artificial Wetland: 1,715,074
Xinhai Artificial Wetland: 1,712,591
Fuzhou Artificial Wetland: 1,422,190
Daniaopi Artificial Wetland: 1,308,525
Lujiaoxi Artificial Wetland: 1,184,552
Chenglin Artificial Wetland: 1,123,061
Xinhai Phase III Artificial Wetland: 1,028,960
Huajiang Artificial Wetland: 720,382

In summary, this study concludes that while all constructed wetlands along the Dahan River are located in urbanized environments, the land use structures surrounding each wetland are inconsistent, resulting in significant differences in the relative environmental pressure they experience. In other words, this study does not present absolute pollution values for each wetland, but rather the relative spatial pressure faced by different constructed wetlands under current land use conditions. These results can serve as a reference for subsequent management, reinforcement, and spatial planning of constructed wetlands.

7. REFERENCE

1. [新北市都市計畫土地使用分區及範圍圖](#)
2. [鄉鎮市區界線\(TWD97經緯度\)](#)
3. [河川流域範圍圖](#)
4. 濕地座標:google map
5. [濕地設立目的](#)